On the (In)significance of Data Validity in Individual Fairness Testing

Authors are blinded for double blind review

Abstract—**Background: Individual fairness testing identifies individual discriminatory instances (IDIs) in a classifier, enabling fairness enhancement through model retraining. Recently, researchers have argued for incorporating "*data validity*", i.e., domain-specific feature constraints inherent in datasets, into fairness testing.**

Aims: This study presents an empirical investigation into the significance of IDI validity. Specifically, we examine whether the validity of identified IDIs systematically influences retraining outcomes in terms of fairness improvement and accuracy preservation.

Method: We develop REVA, an empirical evaluation framework to assess the impact of data validity of remediation of classifiers. Using REVA, we conduct an empirical study across multiple datasets. For experimental data obtained from REVA, we employ partial correlation analysis to isolate the effects of IDI validity while controlling for the confounding influence of IDI diversity.

Results: Our results demonstrate that IDI validity does not exhibit a consistent or statistically significant correlation with either fairness improvement or accuracy degradation in most scenarios. While pronounced effects occur in a few cases, the impact's direction and magnitude vary substantially across tasks.

Conclusions: These findings indicate that IDI validity is not a robust or meaningful standalone metric for evaluating fairness testing algorithms. Because its impact is highly unstable and scenario-dependent, validity alone should not be adopted as a standard evaluation criterion.

Index Terms—**Fairness testing, Empirical Study, Evaluation metrics**

I. INTRODUCTION

Individual Fairness Testing (IFT), introduced by Galhotra et al. [1], is a machine learning (ML) fairness testing framework designed for tabular data. The framework has two main objectives: (1) to identify "*individual discriminatory instances (IDIs)*" in a given classifier under test (CuT), and (2) to improve the fairness of the CuT by retraining it using the identified IDIs. The framework defines an IDI as a pair of data instances that are identical except for their *protected attributes* but receive different outcomes from the classifier. Protected attributes are those that should not be considered in the decision-making process. Table I presents an example of IDIs, adapted from [2].

Research efforts in the fairness testing literature have primarily focused on algorithm development, and consequently, numerous IFT algorithms have been proposed to date, including [1], [4]–[20]. Evaluation of IFT algorithms have been established as well, in the course of it. IFT algorithms are typically evaluated using two primary metrics (as such evaluated in [6] and as clarified [19]): algorithmic efficiency and the performance of the retrained classifiers in terms of fairness and accuracy. Algorithmic efficiency captures how many IDIs an IFT algorithm can identify within a given time constraint. The performance of the retrained classifier metrics evaluate the extent to which the algorithm enhances the fairness of the CuT after retraining, while also accounting for any potential degradation in accuracy. It is worth noting that fairness and accuracy often exhibit a trade-off, adding complexity to the evaluation process.

A recent study highlights the necessity of incorporating *data validity* or *data reality* into individual fairness testing frameworks [2]. According to the study, data validity refers to whether discriminatory instances conform to the semantic constraints inherent among the dataset features. For instance, the aforementioned two individual instances (I_1 and I_2) lack realism in the following aspects, rendering them invalid:

1) The gender of individual I_1 is specified as 'Female', yet 'Relation' is 'Husband,' a designation incongruent with the conventional definition of 'Husband', following a standard definition that denotes "the man that someone is married to" [21].
2) The relationship of individual I_1 (and I_2) is set to 'Husband', although he is divorced.
3) I_1 and I_2, both 17 years old and Japanese citizens, are divorced, while the legal age for marriage in Japan is 18.

To address this issue, the study proposed a new individual fairness testing framework that accounts for data validity, called IFT-V, together with its implementation, IFT-V$_{t-way}$, in which data validity are derived from the original training dataset (described in Section II-B). The study demonstrates that, on average, only a small portion of IDIs detected by a SoTA IFT algorithm are valid, with preliminary experiments, using the framework.

The authors further argue that incorporating data validity ensures that the IDIs used for retraining are realistic and closely aligned with the original data distribution [2]. Such alignment is hypothesized to mitigate accuracy degradation during retraining while preserving fairness. However, this fundamental hypothesis has not yet been validated either empirically or theoretically. Furthermore, the authors suggest that incorporating data validity represents the next paradigm in IFT, making it an indispensable component of future IFT algorithms. Conversely, accounting for data validity introduces non-trivial computational complexity into IFT frameworks,

TABLE I

AN EXAMPLE OF INDIVIDUAL DISCRIMINATORY INSTANCES (IDIS), SIMPLIFIED FROM THE ADULT DATASET [3] FOR PREDICTING WHETHER AN INDIVIDUAL'S INCOME EXCEEDS 50K USD. THE PAIR OF DATA INSTANCES, I_1 AND I_2, REPRESENTS TWO INDIVIDUALS CHARACTERIZED BY SIX FEATURES (E.G., Age, Education, AND Gender), WITH THE FINAL COLUMN INDICATING THE CLASSIFICATION OUTCOME. ASSUMING GenderIS THE PROTECTED ATTRIBUTE, THESE INDIVIDUALS DIFFER SOLELY IN THIS FEATURE YET RECEIVE DIFFERENT PREDICTIONS FROM THE CLASSIFIER.

Attr.	Age	Education	Gender	Relation	Marital Status	Country	>50K?
I_1	17	Bachelor	Male	Husband	Divorced	Japan	Yes
I_2	17	Bachelor	Female	Husband	Divorced	Japan	No

particularly because it requires reasoning over validity constraints that are typically expressed as complex logical formulas.

In this paper, we empirically investigate the necessity of incorporating data validity into individual fairness testing (IFT) frameworks, guided by the following fundamental question:

Does data validity matter in individual fairness testing?

Since our investigation focuses on the remediation aspect of individual fairness testing, we reformulate this question into a more concrete and measurable research question:

RQ: Does the data validity of IDIs affect the fairness improvement and accuracy degradation of retrained classifiers?

An affirmative answer to this research question would provide strong empirical evidence supporting the hypothesis that incorporating data validity improves the effectiveness of IFT-based retraining.

The main contributions of this paper are as follows:

1) We develop an empirical framework, called REVA, for analyzing the effect of data validity on improvements in fairness and degradation in accuracy after classifier retraining. The framework is built upon IFT-V and IFT-V$_{t-way}$ proposed by Kitamura et al. [2]. Specifically, REVA enables quantitative analysis of this effect using the Pearson correlation coefficient (PCC).
2) We conduct extensive experiments using REVA on real-world datasets to empirically investigate the effect of data validity on retraining performance. Our evaluation provides a comprehensive empirical analysis from multiple perspectives, including (1) PCC analysis using the conventional fairness metric (IFr) [1], [4], (2) analysis using a data-validity-aware fairness metric, referred to as valid-IFr (Section III-B2), (3) analysis of the effect of accounting for data validity on data diversity (Section IV-D), and (4) PCC analysis that accounts for data diversity (Section IV-E).

This paper is organized as follows. Section II reviews key concepts, including the IFT framework, data validity, and related metrics. Section III-A presents the REVA framework used to systematically analyze the effect of IDI validity on retraining outcomes. Section IV reports the experimental results and analyzes the relationship between IDI validity and retraining performance. Section VI reviews existing studies on IDI quality metrics and proxy metrics in software testing. Section V discusses threats to validity. Finally, Section VII concludes the paper and outlines directions for future work.

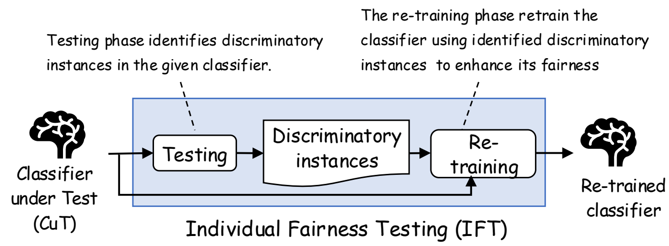

Fig. 1. Overview of Individual Fairness Testing (IFT). IFT consists of two phases: the testing and retraining phases.

A replication package of our work (including code, dataset, and experimental results) is made available at: https://github.com/anon-user-26/Validity_Fairness_Testing

II. PRELIMINARIES

A. Individual Fairness Testing

Individual Fairness Testing (IFT) is a prominent ML fairness testing paradigm, pioneered by Galhotra et al. [1] and subsequently extended by numerous studies (e.g., [4]–[20]). The primary objective of IFT is twofold: to identify IDIs for a given classifier (CuT) and subsequently retrain the CuT using the identified instances to improve its fairness. Accordingly, the workflow typically consists of two distinct phases, as illustrated in Figure 1.

During the retraining phase, classifier fairness and predictive accuracy are widely recognized to exhibit a trade-off. This trade-off arises because synthesized IDIs often deviate from the original training data distribution. Since maximizing fairness while minimizing accuracy degradation remains a challenging problem, prior studies have explored various mitigation strategies. For example, some methods limit the number of synthesized instances injected during retraining to 10% of the original training set size [4], whereas others prioritize highly diverse instances for retraining [15], [19]. However, these strategies focus on the quantity or diversity of synthesized instances, rather than their semantic validity.

B. Individual Fairness Testing with Data Validity: IFT-V

Kitamura et al. [2] proposed IFT-V, a novel paradigm for incorporating data validity into IFT frameworks. An overview of the IFT-V architecture is illustrated in Figure 1. The core workflow of IFT-V consists of the following two phases:

1) **Retrieval of Validity Constraints:** Extracting data validity constraints from the training dataset.

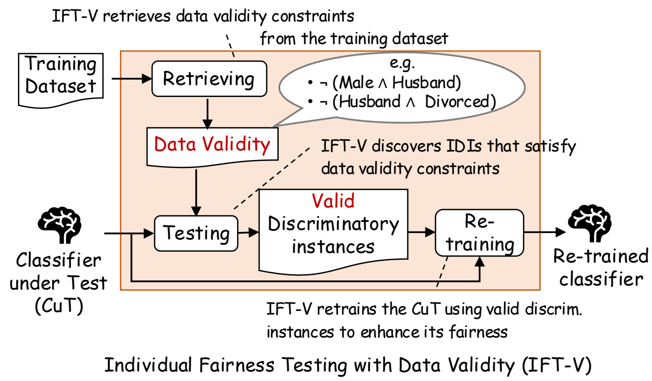

Fig. 2. Overview of Individual Fairness Testing with Data Validity (IFT-V). IFT-V extends IFT with the concept of data validity (Section II-B).

2) **Valid IDI Identification:** Generating and filtering individual discriminatory instances (IDIs) that satisfy the retrieved constraints.

1) Retrieval of Validity Constraints: The IFT-V framework extracts data validity constraints directly from the training dataset. Specifically, for each attribute-value combination observed in the input space, IFT-V classifies the combination as invalid if its empirical frequency in the training data is below a predefined threshold θ; otherwise, the combination is regarded as valid. For example, the combination 'Gender= Female" and 'Relation= Husband" is classified as invalid if its empirical frequency in the training dataset is less than θ.

An extracted invalid combination, denoted by τ, is transformed into a logical validity constraint ϕ as follows:

$$\phi = \neg \left(\bigwedge_{(\text{attr},\text{value}) \in \tau} (\text{attr} = \text{value}) \right)$$

For example, suppose that the frequency of the value combination $\tau_1 = (\text{Gender}, \text{Female}), (\text{Relation}, \text{Husband})$ is less than θ. The following validity constraint is then extracted:

$$\phi_{\tau_1} = \neg(\mathsf{Gender} = \mathsf{Female} \wedge \mathsf{Relation} = \mathsf{Husband})$$

Similarly, the other two constraints introduced in Section I are extracted as:

$$\phi_{\tau_2} = \neg(\mathsf{Relation} = \mathsf{Husband} \wedge \mathsf{Marital} = \mathsf{Divorced})$$
$$\phi_{\tau_3} = \neg(\mathsf{Age} < 18 \ \wedge \ \mathsf{Country} = \mathsf{Japan} \wedge \mathsf{Marital} = \mathsf{Divorced})$$

Notably, this IFT-V framework is primarily conceptual rather than computationally practical, because deriving all possible validity constraints requires examining every possible attribute-value combination. This results in a combinatorial explosion of $O(v^k)$ combinations, where k is the number of attributes and v is the average number of values per attribute.

To address this scalability issue, Kitamura et al. [2] also proposed IFT-V$_{t-way}$, a practical implementation of the IFT-V framework. IFT-V$_{t-way}$ introduces the notion of *combination size*, defined as the number of attribute-value pairs in a combination. For example, the sizes of τ_1, τ_2, and τ_3 are two, two, and three, respectively. Using this notion, IFT-V$_{t-way}$ limits constraint extraction to combinations whose size does not exceed a user-specified threshold s. For example, when $s = 2$, only ϕ_{τ_1} and ϕ_{τ_2} are extracted, thereby substantially reducing the computational cost. This design is inspired by combinatorial t-way testing [22], [23], enabling predictable control of the search space by bounding the maximum combination size.

2) Valid IDI Identification: This phase identifies IDIs that satisfy the extracted data validity constraints. Formally, given a set of retrieved constraints Φ, the objective is to identify an IDI I that satisfies every constraint in the set of derived data validity constraints Φ, denoted by

$$I \models \bigwedge_{\phi \in \Phi} \phi.$$

To achieve this objective, the implementation of IFT-V$_{t-way}$ adopts a generate-and-filter strategy. Specifically, it first generates candidate IDIs using an existing IFT algorithm (EXPGA [9] in their implementation) while temporarily ignoring data validity, and then filters out the candidates that violate the extracted validity constraints.

C. Data Diversity Metric

The diversity of a dataset measures the spatial dispersion of its instances within the input space. Following established studies [15], [16], [19], the diversity of a dataset is quantified as the average pairwise distance among all instances, formally defined as follows.

Definition 1 (Dataset Diversity): Let $\mathcal{D} = I_1, \ldots, I_m$ be a dataset consisting of m instances. The diversity score of $\mathcal{D}$, denoted by $\text{DIV}(\mathcal{D})$, is defined as

$$\text{DIV}(\mathcal{D}) := \frac{\sum_{1 \le i < j \le m} d(I_i, I_j)}{\binom{m}{2}}, \qquad (1)$$

where $d(I_i, I_j)$ denotes the distance between instances I_i and I_j.

This formulation is agnostic to the choice of distance metric $d(\cdot, \cdot)$, allowing any application-specific distance function to be used. Existing fairness testing studies typically adopt either the L_0 or L_1 norm. However, the L_0 norm (e.g., [9], [15], [16]) does not capture the magnitude of differences between numerical attribute values, although it is well suited to purely categorical data. Consequently, recent work [19] has adopted the L_1 norm, which accounts for the magnitude of attribute differences and therefore provides a more faithful representation of spatial dispersion in mixed-type tabular datasets containing both numerical and categorical features.

III. REVA AND METRICS

In this section, we propose our empirical framework, called REVA, for analyzing significance of data validity in fairness testing. We also propose several evaluation metrics to be used for our evaluation in Section IV.

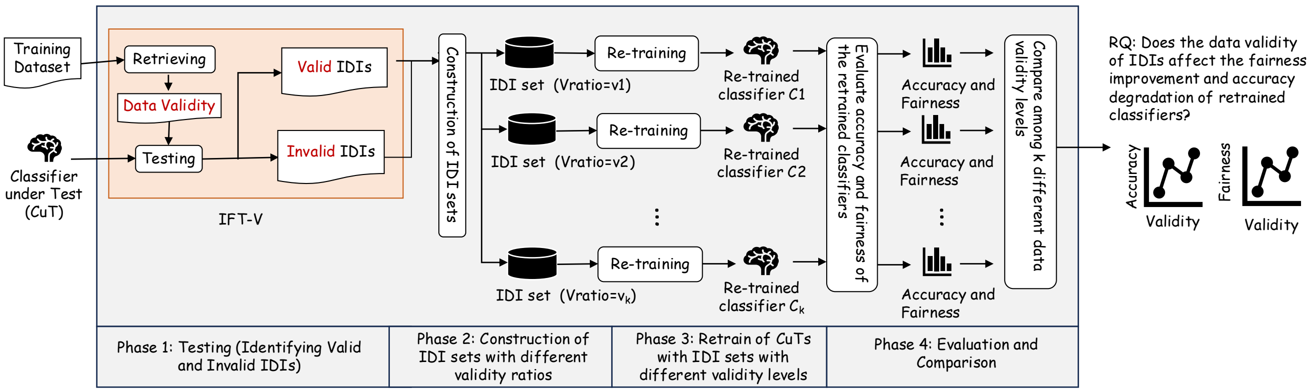

Fig. 3. Overview of the empirical framework REVA.

A. REVA: A Empirical Framework for Analyzing Significance of Data Validity in Fairness Testing

The overview of REVA framework is illustrated in Figure 3. REVA consists of four phases: (1) Testing to identify valid and invalid IDI, (2) validity-controlled IDI sets construction, (3) retraining of CuTs, and (4) evaluation and statistical analysis.

1) Phase-1: Testing to identify valid and invalid IDI: The objective of Phase 1 is to find large sets of valid and invalid IDIs, where we desire both sets large so as to be used in the later phases. For this objective, REVA utilized the framework IFT-V, more specifically IFT-V$_{t-way}$explained in Section II-B [2]. Recall that IFT-V$_{t-way}$ is designed to identify valid IDIs with respect to up to the user-specified validity constraint size s. The mechanism adopts a naive generate-filter approach, which first identifies IDIs disregarding the data validity using any existing IFT algorithms, and filter them according to data validity constraints.

We implement IFT-V$_{t-way}$ by ourself, since the the source code is not available from [2]. For the implementation, we employ AFT [16] for the IFT algorithm used in IFT-V$_{t-way}$, since the algorithm is SoTA IFT algorithm that can efficiently generate diverse IDIs. The algorithm is desirable for our purpose since we need a need a huge number of IDIs and IDIs diversity is also important.

To instantiate the validity definition, we adopt parameter settings based on prior work [2], which provide a practical trade-off between expressiveness and computational cost. Specifically, we set the interaction strength to $t = 2$ and the frequency threshold to $\theta = 0$, following [2], while the number of bins is set to $k = 5$. Setting $\theta = 0$ implies that any interaction not observed in the training dataset is treated as invalid. This configuration corresponds to a strict validity definition and is commonly used in prior studies. For numerical attributes, values are discretized into $k = 5$ bins before extracting t-way interactions. While prior work uses $k = 10$, we reduce this value to increase the number of valid IDIs detected, which is necessary for constructing sufficient retraining datasets within the REVA framework.

The set $\mathrm{IDI}_{\mathrm{valid}}$ corresponds to discriminatory behaviors on realistic data instances that satisfy empirical data-validity constraints, whereas $\mathrm{IDI}_{\mathrm{invalid}}$ arise from violations of these constraints and are discriminatory behaviors on unrealistic instances.

We run AFT for a maximum duration of one hour. The execution is terminated earlier if a sufficient number of valid and invalid IDIs required for retraining are obtained.

2) Phase-2: Validity-controlled IDI sets construction: To analyze the effect of IDI validity on retraining outcomes, Phase 3 constructs IDI sets with systematically controlled proportions of valid and invalid IDIs.

Let n denote the fixed number of IDIs used for retraining. For a given validity ratio $\mathrm{Vratio} \in [0, 1]$, we construct a retraining IDI set as:

Definition 2:

$$\begin{aligned}\mathrm{IDI}_{\mathrm{retrain}}(\mathrm{Vratio}) = \mathrm{Sample}(\mathrm{IDI}_{\mathrm{valid}}, \mathrm{Vratio} \cdot n) \\ \cup\ \mathrm{Sample}(\mathrm{IDI}_{\mathrm{invalid}}, (1 - \mathrm{Vratio}) \cdot n)\end{aligned} \tag{2}$$

Here, Vratio denotes the proportion of valid IDIs in the retraining set.

To systematically vary this proportion, we introduce a hyperparameter K, which specifies the number of validity levels. We discretize the interval $[0, 1]$ into K evenly spaced values:

$$\mathrm{Vratio} \in \left\{ \frac{i}{K-1} \,\middle|\, i = 0, 1, \ldots, K-1 \right\}. \tag{3}$$

This results in K retraining IDI sets with different validity ratios.

In this study, we set $K = 21$, which provides a sufficiently fine-grained range to reliably estimate correlations, while keeping the computational cost manageable.

In prior work, validity is defined at the instance level, i.e., each IDI is classified as either valid or invalid based on data-validity constraints. However, no metric has been proposed to quantify the validity of an IDI set as a whole. Following existing instance-level definitions, we define the validity of an IDI set as the proportion of valid IDIs it contains, i.e., Vratio.

3) Phase4: Retraining of CuT: In Phase 4, each IDI set constructed in Phase 3 is used to retrain the CuT. Specifically, all 21 IDI sets with different levels of validity are each used for retraining, resulting in 21 retrained classifiers. The retraining procedure follows the method introduced by Funamoto et al. [19] . That is, for retraining of each IDI, we use both instances in the discriminatory pair and assign their labels by majority voting. From an evidence-based perspective, this retraining strategy has been shown to yield stable fairness improvement, making it a suitable choice for our study.

4) Phase5: Evaluation and Comparison: To answer RQ, we examine whether IDI validity is correlated with the retraining outcomes while removing the confounding influence of IDI diversity. Since validity and diversity are inherently related, a simple Pearson correlation cannot separate their respective effects. For each scenario, we compute the partial correlation between IDI validity and (i) accuracy improvement, (ii) IFr improvement, and (iii) Valid-IFr improvement, while controlling for diversity (measured by Binned L_0-distance). This analysis reveals whether validity itself—independent of diversity—is associated with changes in classifier performance after retraining.

B. Metrics for Dataset Diversity and Classifier Fairness

1) Diversity measure of IDIs: As discussed in Section II-C, existing fairness-testing studies typically compute distances between data instances using either the L_0 or L_1 norm. The L_0 norm is traditionally adopted due to its simplicity but cannot faithfully reflect differences among instances because it treats all attribute mismatches uniformly, regardless of their magnitude. The L_1 norm can avoid this limitation by incorporating the size of numeric differences, it is inappropriate for categorical attributes, since numeric subtraction is semantically meaningless.

To address these issues, we introduce a new distance metric, *Binned L_0-distance*. The Binned L_0-distance first discretizes all numeric attributes into fixed-width bins and then computes distance based on the $L-0$ norm over the resulting categorical representations. This approach (i) avoids inappropriate numeric comparisons on categorical attributes, and (ii) provides a more faithful notion of proximity, since instances falling into the same bin are treated as identical while those separated by wider numeric gaps fall into different bins. In this work, *Binned L_0-distance* serves as the underlying distance metric for computing diversity scores in all subsequent analyses.

2) Fairness of Classifiers: To evaluate the fairness of classifiers, we employ two metrics: the Individual Fairness Ratio ($\mathrm{IF_r}$) and its validity-aware variant, Valid-$\mathrm{IF_r}$.

a) Individual Fairness Ratio (IFr): The Individual Fairness Ratio ($\mathrm{IF_r}$) is a widely used metric in fairness testing, which estimates the prevalence of IDIs in the input space. The idea is to sample a large number of test instances and measure the proportion that exhibit discriminatory behavior.

Definition 3 (Individual Fairness Ratio (IFr)): Let $\#S_{idi}$ denote the number of detected IDIs and let $|S|$ represent the total number of generated test instances. The individual fairness ratio is defined as:

$$\mathrm{IFr} := \frac{\#S_{idi}}{|S|}. \tag{4}$$

A smaller value of IFr indicates that fewer discriminatory instances are found within the sampled input space, implying that the classifier exhibits higher individual fairness.

Algorithm 1 Validity-aware Fairness Metric (Valid-IFr)

Require: f: Classifier under test, $\mathcal{D}$: Input domain, A_p: Protected attribute, Domain of A_p, Validity constraints, Number of valid trials

$count_{\text{valid\_idi}} \leftarrow 0$
for $i \leftarrow 1$ to N **do**
  while true **do**
    $x \leftarrow$ Randomly generate an instance from $\mathcal{D}$
    $valid\_instances \leftarrow \emptyset$
    for all $v \in V_p$ **do**
      $x' \leftarrow x$ with $A_p = v$
      if $\mathrm{IsValid}(x', \Phi)$ **then**
        Append x' to $valid\_instances$
    if $|valid\_instances| \geq 2$ **then**
      break ▷ Ensure comparability across values
  $outputs \leftarrow \{f(x') \mid x' \in valid\_instances\}$
  if not all values in $outputs$ are identical **then**
    $count_{\text{valid\_idi}} \leftarrow count_{\text{valid\_idi}} + 1$
return valid-$\mathrm{IF_r} = \frac{count_{\text{valid\_idi}}}{N}$

b) Validity-aware Fairness Metric (Valid-IFr): In addition to $\mathrm{IF_r}$, we introduce a refined fairness metric, termed Valid-$\mathrm{IF_r}$. While $\mathrm{IF_r}$ estimates the overall prevalence of discriminatory behavior across the full combinatorial input space, Valid-$\mathrm{IF_r}$ focuses exclusively on the valid region of the input space, i.e., instances whose attribute-value combinations satisfy the data-validity constraints derived from the training data.

Formally, let S_{valid} denote a set of randomly generated test instances that are judged valid with respect to data validity constraints derived from the training data, and let $\#S_{\text{valid\_IDI}}$ be the number of IDIs among them. The Valid Individual Fairness Ratio (Valid-IFr) is defined as follows:

$$\text{Valid-IF}_{\mathrm{r}} = \frac{\#S_{\text{valid\_IDI}}}{|S_{\text{valid}}|} \tag{5}$$

Valid-$\mathrm{IF_r}$ captures the likelihood that a deployed classifier exhibits discriminatory behavior on realistic and plausible inputs, rather than on arbitrary points in the entire input space. Since practical fairness concerns predominantly arise in this valid data region, we evaluate retrained classifiers using both $\mathrm{IF_r}$ and the proposed Valid-$\mathrm{IF_r}$.

IV. Experiments

In this section, we present the empirical results of our evaluation.

TABLE II
DETAILS OF DATASETS. THE ‘SIZE’, ‘#ATTR’, AND ‘PROTECTED ATTR.’ COLUMNS MEAN THE NUMBER OF DATA INSTANCES, THE NUMBER OF ATTRIBUTES, AND THE CANDIDATES OF PROTECTED ATTRIBUTES.

Dataset	Size	#Attr.	Protected Attr.
Census Income [3] (ADULT)	32561	13	Gender, Race, Age
Bank Marketing [24] (BANK)	45211	16	Age
German Credit Data [25] (CREDIT)	1000	20	Gender, Age

TABLE III
ML CLASSIFIERS AND THEIR TRAINING PARAMETERS.

Classifier	Parameters
Support Vector Machine (SVM)	$\gamma = 0.0025$
Random Forest (RF)	criterion=’gini’, 10 trees, max_depth=5
Multilayer Perceptron (MLPC)	One hidden layer (100 units), ReLU, Adam

TABLE IV
SUMMARY OF THE EFFECTS OF VALIDITY ON FAIRNESS AND ACCURACY IMPROVEMENTS (SIMPLE CORRELATION ANALYSIS)

				Fair. ($\mathrm{IF_r}$)		Fair. (valid-$\mathrm{IF_r}$)		Acc.	
No.	Clf	Data	p.a.	PCC_p	p-value	PCC_p	p-value	PCC_p	p-value
1	SVM	Adult	Age	**-1.0**	**<0.01**	**-0.93**	**<0.01**	**-0.96**	**<0.01**
2		Credit	Age	**-0.81**	**<0.01**	**-0.76**	**<0.01**	0.29	0.20
3			Gender	**0.47**	**0.03**	**0.58**	**<0.01**	**0.49**	**0.03**
4	MLPC	Adult	Age	-0.25	0.27	0.38	0.09	-0.22	0.35
5			Race	0.04	0.87	**0.50**	**0.02**	-0.01	0.95
6			Gender	0.26	0.26	**0.54**	**0.01**	-0.03	0.90
7		Bank	Age	0.03	0.90	0.17	0.47	-0.08	0.72
8		Credit	Age	0.11	0.62	0.16	0.49	0.21	0.36
9			Gender	0.01	0.96	-0.03	0.88	0.17	0.46
10	RF	Bank	Age	-0.36	0.11	**0.90**	**<0.01**	0.14	0.56
11		Credit	Age	**-0.47**	**0.03**	-0.00	0.99	0.01	0.97
12			Gender	**-0.88**	**<0.01**	-0.16	0.49	-0.11	0.64
				-0.24	5	0.11	6	-0.01	2

A. Research Questions

In order to assess the significance of IDI data validity in retraining, we primarily analyze the correlation between the data validity ratio of IDIs and the resulting retraining outcomes, specifically fairness improvement and accuracy degradation. To ensure a rigorous analysis, we supplement the simple correlation study with a more detailed investigation. Specifically, we examine the potential influence of dataset diversity on data validity and subsequently isolate this factor using partial correlation analysis. This methodical approach leads us to formulate the following concrete research questions (RQs):

RQ1: Is the validity ratio of IDIs correlated with the fairness improvement and accuracy degradation of retrained classifiers under a simple correlation analysis?

RQ2: Does the validity ratio of IDIs affect the dataset diversity?

RQ3: Is the validity ratio of IDIs correlated with the fairness improvement and accuracy degradation of retrained classifiers when controlling for dataset diversity via partial correlation analysis?

B. Datasets and Classifiers

Our experiments draw on three widely used datasets and three standard classifiers, summarized in Table II and Table III. Consistent with common practice, we create fairness-testing tasks by pairing each dataset (together with its designated protected attribute) with each classifier, yielding 18 distinct tasks (see Table V or Table IV). For the ADULT dataset, we follow earlier work [4], [17] and restrict the data to the 32,561 training instances supplied in the UCI split, rather than using the full 48,842-instance corpus.

C. RQ1: Is the validity ratio of IDIs correlated with the fairness improvement and accuracy degradation of retrained classifiers under a simple correlation analysis?

Table IV summarizes the evaluation results across the 12 tasks. The analysis was restricted to these tasks because the REVA framework requires a sufficient pool of valid IDIs to construct retraining datasets. In 6 of the original 18 tasks, AFT failed to detect the minimum number of valid IDIs required for retraining; thus, they were excluded from the RQ analysis.

For fairness improvement (measured by $\mathrm{IF_r}$), the average Pearson correlation coefficient (PCC) is -0.24, indicating a weak negative correlation overall. No statistically significant correlation is observed at the 5% significance level in 7 of the 12 scenarios. Significant negative correlations exist in 4 scenarios (No. 1, No. 2, No. 10, and No. 12), while a positive correlation is found in only one (No. 3). This highlights a strong scenario dependency, suggesting that IDI validity is largely uncorrelated with fairness improvement except in specific scenarios (i. e., dataset and protected attribute).

For valid-$\mathrm{IF_r}$ improvement, the average PCC is 0.11, indicating no overall correlation. While 6 of the 12 scenarios show no statistically significant correlation, the remaining 6 exhibit significant correlation: No. 1 and No. 2 show strong negative correlations, whereas No. 3, No. 5, No. 6, and No. 9 show positive correlations, with No. 9 being exceptionally strong (PCC $= 0.90$). This indicates that while there is no systematic trend, individual scenarios can exhibit polarized effects depending on scenarios.

Regarding accuracy degradation, the average PCC across the 12 tasks is -0.01, showing a negligible overall correlation. In 10 of the 12 scenarios, no statistically significant correlation is observed. A significant negative correlation appears only in No. 1, and a positive one in No. 3. These results suggest that IDI validity generally exerts little influence on accuracy degradation, though localized effects may manifest.

Answer to RQ1: Our simple correlation analysis reveals no systematic correlation between the IDI validity ratio and retraining outcomes (average PCC ≈ 0 across fairness and accuracy metrics for most tasks). While statistically insignificant in the majority of scenarios, the relationship is highly scenario-dependent, occasionally yielding pronounced, polarized effects in specific scenarios.

D. RQ2: Does the validity ratio of IDIs affect their dataset diversity?

Table V summarizes the PCCs and corresponding p-values between validity and diversity for the 21 IDI sets generated

TABLE V

CORRELATIOIN COEFFICIENT BETWEEN VALIDITY AND DIVERSITY

No.	Clf	Data	p.a.	Validity vs. Diversity PCC	p-value
1	SVM	Adult	Age	-0.94	<0.01
2		Credit	Age	-0.94	<0.01
3			Gender	-0.95	<0.01
4	MLPC	Adult	Age	-0.92	<0.01
5			Race	-0.97	<0.01
6			Gender	-0.98	<0.01
7		Bank	Age	-0.95	<0.01
8		Credit	Age	-0.93	<0.01
9			Gender	-0.92	<0.01
10	RF	Bank	Age	-0.94	<0.01
11		Credit	Age	-0.94	<0.01
12			Gender	-0.94	<0.01
				-0.94	12

TABLE VI

SUMMARY OF THE EFFECTS OF DATA VALIDITY ON FAIRNESS AND ACCURACY IMPROVEMENTS ACCOUNTING FOR DATA DIVERSITY (PARTIAL CORRELATION ANALYSIS)

No.	Clf	Data	p.a.	Fair. (IF_r) PCC_p	p-value	Fair. (valid-IF_r) PCC_p	p-value	Acc. PCC_p	p-value
1	SVM	Adult	Age	**-0.98**	**<0.01**	**-0.98**	**<0.01**	**-0.51**	**0.02**
2		Credit	Age	**-0.40**	0.08	**-0.41**	0.07	**0.47**	**0.04**
3			Gender	0.36	0.11	**0.42**	0.07	-0.01	0.97
4	MLPC	Adult	Age	-0.20	0.41	0.22	0.35	0.02	0.94
5			Race	-0.17	0.46	-0.02	0.94	-0.17	0.46
6			Gender	-0.03	0.89	0.29	0.22	0.31	0.19
7		Bank	Age	-0.05	0.84	0.10	0.68	-0.15	0.54
8		Credit	Age	-0.01	0.95	0.09	0.71	**0.47**	**0.04**
9			Gender	0.01	0.97	-0.05	0.84	-0.04	0.87
10	RF	Bank	Age	-0.11	0.64	**0.92**	**<0.01**	0.27	0.24
11		Credit	Age	0.02	0.94	-0.11	0.65	-0.23	0.32
12			Gender	-0.34	0.14	-0.08	0.74	-0.22	0.36
				-0.16	1	0.03	2	0.06	3

by REVA across the 12 scenarios. Notably, every scenario exhibits a strong and statistically significant negative correlation between these two properties. This consistent trend indicates that increasing the validity of IDIs systematically leads to a reduction in the diversity of the generated IDI sets.

This result is intuitively plausible and aligns with expectations. Higher validity requires IDIs to satisfy stringent, domain-specific constraints, which inherently constrains the feasible region within the input space. Consequently, the synthesized IDIs concentrate within narrower regions of the feature space, leading to diminished diversity.

Answer to RQ2: Yes, the validity ratio of IDIs consistently and negatively affects dataset diversity, which aligns with our expectations. Across all 12 scenarios, a strong and statistically significant negative correlation was observed. This finding suggests that to isolate and accurately assess the independent effect of validity on retraining performance, it is crucial to control for the influence of data diversity.

E. RQ3: Is the validity of IDIs correlated with fairness improvement and accuracy degradation of the retrained classifiers?

Table VI summarizes the partial Pearson correlation coefficients (partial-PCCs) across the 12 tasks after controlling for dataset diversity.

For fairness improvement (IF_r), the average partial-PCC is -0.16, indicating a weak negative correlation overall. Out of the 12 scenarios, 11 scenarios exhibit no statistically significant correlation ($p \geq 0.05$). The sole exception is No. 1, which shows a prominent negative correlation (partial-PCC $= -0.98$), reinforcing its strong scenario dependency.

Regarding valid-IF_r improvement, the average partial-PCC is 0.03, confirming no general correlation. Out of the 12 scenarios show no statistically significant correlation. Significant correlations emerge in only two tasks: No. 1 (strong negative; partial-PCC $= -0.98$) and No. 10 (strong positive; partial-PCC $= 0.92$), highlighting scenario-specific polarization.

For accuracy degradation, the average partial-PCC is 0.06, showing a negligible overall correlation. Out of the 12 scenarios, 9 show no statistically significant correlation. Although significant correlations occur in No. 1, No. 2, and No. 8, their moderate values (-0.51, 0.47, and 0.47, respectively) suggest that IDI validity generally exerts minimal influence on accuracy.

Table VII details four representative tasks illustrating how the 21 controlled data validity levels of IDIs affect retraining outcomes when data diversity is accounted for. The complete results are available in our GitHub repository.

Tasks 2 and 9 represent cases where IDI validity has no statistically significant correlation with any retraining outcome. In Task 9, all absolute partial PCCs are below 0.1, indicating that validity plays virtually no role in this scenario.Conversely, Task 1 exhibits a case where a clear negative trend manifests: as the data validity level increases, both IF_r and valid-IF_r improvements systematically decrease. Notably, when the validity level exceeds 0.9, IF_r improvement becomes negative, indicating that retraining with highly valid IDIs can actively deteriorate fairness. This highlights that higher validity does not inherently guarantee better fairness outcomes and can occasionally yield adverse effects.

Task 10 exhibits a case where retraining improves fairness in terms of valid-IF_r without degrading accuracy. This case also provides an interesting phenomenon: while the IF_r improvement remains almost unchanged as the validity level increases, the valid-IF_r improvement shows a clear increasing trend. This indicates that improvements measured by IF_r and valid-IF_r do not necessarily align and can respond differently to changes in data validity.

TABLE VII

IFR, VALID-IFR, AND ACCURACY IMPROVEMENTS BEFORE AND AFTER RETRAINING WITH IDIS OF VARYING VALIDITY FOR SELECTED TASKS

1. SVM-Adult-Age

		Before			After			Impr. (%)		
Val.	Div.	IFr	valid-IFr	Acc.	IFr	valid-IFr	Acc.	IFr	valid-IFr	Acc.
0.00	9.90				6.82	15.84	80.86	30.11	44.36	-0.12
0.05	9.91				7.13	16.22	80.86	26.91	43.03	-0.11
0.10	9.91				7.34	16.36	80.83	24.79	42.54	-0.15
0.15	9.91				7.46	16.47	80.82	23.63	42.13	-0.16
0.20	9.90				7.61	16.59	80.82	22.09	41.71	-0.16
0.25	9.89				7.69	16.94	80.85	21.23	40.50	-0.13
0.30	9.88				7.98	17.32	80.82	18.29	39.17	-0.16
0.35	9.85				8.15	17.34	80.82	16.54	39.08	-0.16
0.40	9.83				8.22	17.48	80.78	15.75	38.60	-0.21
0.45	9.80				8.34	17.50	80.75	14.59	38.52	-0.25
0.50	9.76	9.76	28.47	80.95	8.53	17.79	80.81	12.61	37.50	-0.18
0.55	9.73				8.78	17.95	80.76	10.05	36.95	-0.23
0.60	9.69				8.75	17.88	80.74	10.41	37.18	-0.26
0.65	9.64				9.04	18.08	80.74	7.42	36.50	-0.26
0.70	9.60				9.13	18.10	80.67	6.52	36.41	-0.35
0.75	9.54				9.31	18.16	80.67	4.67	36.21	-0.35
0.80	9.47				9.43	18.10	80.66	3.38	36.42	-0.36
0.85	9.40				9.48	18.26	80.65	2.87	35.85	-0.37
0.90	9.32				9.81	18.25	80.60	-0.54	35.89	-0.43
0.95	9.25				9.89	18.22	80.61	-1.28	35.99	-0.42
1.00	9.17				9.95	17.91	80.58	-1.97	37.10	-0.46

Fairness (IFr) PCC / Partial-PCC (p-value): -1.0 (< 0.01) / -0.98 (< 0.01)
Fairness (valid-IFr) PCC / Partial-PCC (p-value): -0.93 (< 0.01) / -0.98 (< 0.01)
Accuracy PCC / Partial-PCC (p-value): -0.96 (< 0.01) / -0.51 (0.02)

2. SVM-Credit-age

		Before			After			Impr. (%)		
Val.	Div.	IFr	valid-IFr	Acc.	IFr	valid-IFr	Acc.	IFr	valid-IFr	Acc.
0.00	13.59				6.66	6.17	75.70	-11.40	-21.73	-0.98
0.05	13.46				4.44	4.27	76.15	25.71	15.68	-0.39
0.10	13.50				4.87	4.67	76.00	18.60	7.88	-0.59
0.15	13.56				6.49	5.70	76.10	-8.49	-12.51	-0.46
0.20	13.57				6.47	6.24	75.95	-8.18	-23.11	-0.65
0.25	13.55				6.18	6.06	76.65	-3.33	-19.65	0.26
0.30	13.54				4.92	4.71	76.65	17.73	6.98	0.26
0.35	13.45				5.96	5.75	76.05	0.38	-13.48	-0.52
0.40	13.42				6.78	5.99	76.30	-13.39	-18.18	-0.20
0.45	13.26				6.87	6.13	76.00	-14.95	-20.98	-0.59
0.50	13.35	5.98	5.07	76.45	7.66	6.99	76.15	-28.17	-38.01	-0.39
0.55	13.30				7.48	6.62	76.50	-25.07	-30.64	0.07
0.60	13.19				8.67	7.77	76.15	-44.93	-53.35	-0.39
0.65	13.13				6.71	5.97	76.55	-12.22	-17.81	0.13
0.70	13.07				6.37	5.63	76.35	-6.50	-11.11	-0.13
0.75	12.97				8.26	7.39	76.60	-38.11	-45.92	0.20
0.80	12.85				7.49	6.50	75.70	-25.25	-28.35	-0.98
0.85	12.79				8.57	7.46	76.65	-43.40	-47.33	0.26
0.90	12.67				9.03	7.67	76.35	-51.04	-51.50	-0.13
0.95	12.51				8.28	7.39	76.50	-38.54	-45.99	0.07
1.00	12.39				9.30	7.66	75.95	-55.54	-51.19	-0.65

Fairness (IFr) PCC / Partial-PCC (p-value): -0.81 (< 0.01) / -0.40 (0.08)
Fairness (valid-IFr) PCC / Partial-PCC (p-value): -0.76 (< 0.01) / -0.41 (0.07)
Accuracy PCC / Partial-PCC (p-value): 0.29 (0.20) / 0.47 (0.04)

9. MLPC-Credit-Gender

		Before			After			Impr. (%)		
Val.	Div.	IFr	valid-IFr	Acc.	IFr	valid-IFr	Acc.	IFr	valid-IFr	Acc.
0.00	12.75				0.82	0.85	59.80	58.67	59.52	-18.08
0.05	12.77				0.77	0.69	50.20	61.56	67.07	-31.23
0.10	12.82				1.22	1.01	62.95	38.57	51.65	-13.77
0.15	12.86				0.34	0.39	66.10	82.75	81.14	-9.45
0.20	12.91				0.67	0.80	71.00	66.27	61.61	-2.74
0.25	12.83				0.48	0.39	58.85	75.89	81.39	-19.38
0.30	12.78				0.29	0.32	62.10	85.45	84.77	-14.93
0.35	12.69				0.46	0.50	61.10	76.77	76.24	-16.30
0.40	12.79				0.65	0.73	64.80	67.32	65.17	-11.23
0.45	12.65				0.08	0.12	49.75	95.81	94.15	-31.85
0.50	12.58	1.99	2.09	73.00	1.17	1.38	52.20	41.01	34.06	-28.49
0.55	12.58				0.77	0.78	64.85	61.53	62.87	-11.16
0.60	12.52				0.72	0.60	64.50	63.63	71.50	-11.64
0.65	12.38				2.00	2.04	68.55	-0.54	2.32	-6.10
0.70	12.39				0.46	0.49	50.80	77.07	76.41	-30.41
0.75	12.20				1.02	1.14	53.10	48.81	45.54	-27.26
0.80	12.11				0.49	0.51	57.75	75.27	75.43	-20.89
0.85	12.02				0.72	0.61	62.55	63.81	70.60	-14.32
0.90	11.88				0.46	0.39	70.35	77.02	81.51	-3.63
0.95	11.78				0.47	0.61	63.30	76.27	71.02	-13.29
1.00	11.64				0.61	0.67	70.55	69.52	68.11	-3.36

Fairness (IFr) PCC / Partial-PCC (p-value): 0.01 (0.96) / 0.01 (0.97)
Fairness (valid-IFr) PCC / Partial-PCC (p-value): -0.03 (0.88) / -0.05 (0.84)
Accuracy PCC / Partial-PCC (p-value): 0.17 (0.46) / -0.04 (0.87)

10. RF-Bank-Age

		Before			After			Impr. (%)		
Val.	Div.	IFr	valid-IFr	Acc.	IFr	valid-IFr	Acc.	IFr	valid-IFr	Acc.
0.00	10.55				6.43	7.10	90.29	76.40	65.27	-0.13
0.05	10.57				6.05	6.44	90.35	77.80	68.49	-0.06
0.10	10.57				6.15	6.05	90.41	77.41	70.43	-0.00
0.15	10.57				6.66	6.63	90.42	75.56	67.59	0.02
0.20	10.56				6.62	5.99	90.39	75.72	70.73	-0.02
0.25	10.56				6.58	5.62	90.47	75.86	72.53	0.08
0.30	10.54				5.80	5.51	90.44	78.70	73.08	0.04
0.35	10.52				6.69	5.64	90.44	75.47	72.43	0.03
0.40	10.49				6.78	5.42	90.51	75.12	73.51	0.11
0.45	10.47				6.40	5.39	90.43	76.53	73.64	0.03
0.50	10.43	27.25	20.45	90.41	6.23	5.18	90.36	77.14	74.68	-0.05
0.55	10.39				6.06	5.08	90.38	77.74	75.18	-0.03
0.60	10.35				6.97	5.27	90.36	74.44	74.21	-0.05
0.65	10.30				6.12	4.93	90.35	77.52	75.90	-0.06
0.70	10.24				6.42	4.93	90.34	76.43	75.90	-0.07
0.75	10.19				7.04	5.16	90.40	74.15	74.79	-0.00
0.80	10.14				6.40	4.87	90.47	76.50	76.20	0.07
0.85	10.07				7.02	4.88	90.41	74.26	76.16	0.00
0.90	9.99				6.98	4.89	90.39	74.38	76.09	-0.02
0.95	9.92				6.69	4.89	90.45	75.44	76.09	0.05
1.00	9.84				6.33	4.90	90.41	76.76	76.07	0.00

Fairness (IFr) PCC / Partial-PCC (p-value): -0.36 (0.11) / -0.11 (0.64)
Fairness (valid-IFr) PCC / Partial-PCC (p-value): 0.90 (< 0.01) / 0.92 (< 0.01)
Accuracy PCC / Partial-PCC (p-value): 0.14 (0.56) / 0.27 (0.24)

Answer to RQ3: **Answer to RQ3:** When dataset diversity is accounted for, IDI validity shows no systematic or uniform correlation with retraining outcomes across the majority of tasks. Nonetheless, its influence is highly scenario-dependent and can be substantial; it can occasionally lead to active fairness degradation (Task 1) or cause divergent behaviors between different evaluation metrics (Task 9). Consequently, while validity cannot be generalized as a uniform evaluation metric, its specific impact must be assessed carefully on a task-by-task basis.

F. Answer to the Main Research Question

Based on the insights gained from our empirical investigation of the individual RQs, we conclude this section by addressing our foundational research question as follows:

Should individual fairness testing consider data validity?
Answer: *Yes, but with crucial caveats.* Individual fairness testing frameworks must incorporate data validity not as a uniform, standalone benchmark for algorithm performance, but as a critical, context-dependent safety metric. Since the impact of validity on retraining outcomes is highly scenario-dependent, where effects range from negligible to severe and localized fairness degradation, ignoring it risks unpredictable model behavior during remediation.

V. THREATS TO VALIDITY

As an empirical investigation, this study incurs validity threats from multiple aspects.

a) Definition of Data Validity: Because our findings rely on the definition of data validity operationalized in IFT-V [2], the generalizability of our conclusions is inherently bound to this specific formulation and the choice of hyperparameters. In our empirical evaluation, we utilized fixed values for these parameters, specifically setting the maximum size of attribute-value combinations to $s = 2$, the frequency threshold to $\theta = 0$, and the number of bins to k. Although evaluating a broader configuration space remains imperative to fully establish the external validity of our findings, a comprehensive parameter sweep presents severe tractability challenges due to the combinatorial explosion inherent in the hyperparameter space.

b) Validity Measurement at the Dataset Level: Our proposed framework REVA quantifies the validity of an IDI set using the proportion of valid IDIs, denoted as Vratio Section **??**. Although simple and interpretable, its main limitation is that REVA does not explicitly capture how valid IDIs are distributed in the input space. In this study, we partially mitigate this issue by controlling for diversity when analyzing the relationship between validity and retraining outcomes. Since diversity reflects the dispersion of IDIs in the input space, this approach helps reduce the confounding effect of distributional differences. However, this does not fully eliminate all structural differences among IDI sets, and more expressive metrics may be needed to capture dataset-level validity more precisely.

c) Diversity measures: Our findings rely on the diversity measure based on *Binned L_0-distance*. We selected this metric as an improvement over both the L_0-norm and L_1-norm used in prior studies [15], [16], [19] (as explained in Section III-B1). Exploring alternative diversity measures remains an important direction for future work.

d) IFT *algorithms:* In the experiment, we relied on AFT as the IFT algorithm for generating IDIs. Since different IFT algorithms may produce IDI sets with different characteristics, our findings may depend on this design choice. Investigating whether the same results hold when other IFT algorithms are used is an important direction for future work and would improve the generalizability of our conclusions.

e) Datasets and classification algorithms: Our experiments utilize three widely adopted datasets (ADULT, BANK, and CREDIT), along with three protected attributes (i. e., Age, Gender, and Race) and three classification algorithms (i. e., SVM, MLPC, and RF). While the results of our study are inherently dependent on the specific datasets used, we note that these datasets are standard benchmarks used in the fairness testing community (e. g., [9], [11], [14]–[18]). Therefore, this potential threat to validity is mitigated to a considerable extent by the established use of these datasets in prior work.

VI. RELATED WORK

A. Validity-aware and quality-aware IFT studies

Recent studies in individual fairness testing have begun to emphasize the quality of identified IDIs, introducing concepts such as diversity, naturalness, and validity as desirable properties of IDI sets. Accordingly, several IFT algorithms have been proposed to generate high-quality IDIs: for example, diversity-aware approaches [9], [10], [14], [16] aim to identify IDIs that are broadly distributed in the input space, while validity-aware approaches [2] require IDIs to satisfy logical or semantic constraints inherent in the dataset. The primary focus of these studies is algorithm development, i.e., designing methods to generate IDIs with desirable properties. In contrast, our study asks whether validity is an appropriate evaluation criterion in the first place.

Our work is also inspired by the prior study [19] that investigate whether fairness testing algorithms should consider IDI diversity. The study empirically examines the relationship between IDI diversity and retraining effectiveness. While prior work focuses on diversity, our study extends this line of research to validity.

B. Data Validity Constraints in Tabular ML

Considering data validity constraints within tabular ML is not a novel challenge; rather, ensuring validity has long been a key focus across various domains, most notably in *algorithmic recourse* and *adversarial machine learning*.

Algorithmic recourse [26] is an ML explainability concept that identifies specific, actionable attribute modifications an individual (represented as a tabular data instance) can change to obtain a more favorable prediction from an ML classifier. For example, if an applicant is denied a loan, it prescribes recourse actions, such as increasing savings or clearings debts, to overturn the rejection. To render such recommended recourse actions practically viable, existing literature has intensively investigated data validity constraints to prevent algorithms from proposing impossible or unrealistic attribute modifications (e. g., [27]–[29].

Adversarial attacks on tabular data is a ML security concept where minor perturbations are applied to a data instance to elicit an incorrect or different outcome from an ML classifier. As these attacks have increasingly been studied in the tabular context, Ben-Tov et al. incorporate data validity to craft *feasible and realistic attacks* [30].

Drawing inspiration from these advancements in algorithmic recourse and adversarial ML, this paper presents the first systematic effort to validate the critical yet underexplored importance of data validity specifically in the context of fairness testing.

VII. CONCLUSION AND FUTURE WORK

This paper presents an empirical investigation into the role of data validity within the context of individual fairness testing. Specifically, we examine whether incorporating data validity constraints during the retraining phase mitigates the inherent trade-off between model fairness and predictive accuracy. To facilitate this inquiry, we develop an empirical evaluation framework to assess data validity and perform a detailed analysis of the dataset generated by REVA. In doing so, we carefully disentangle confounding factors of dataset diversity. Our findings is that, contrary to common assumptions in prior literature, from our experiments we can not find enough evidence that validity alone may serve as a viable role in fairness testing; rather, its utility is highly contextual, depending on the intrinsic characteristics of both the data and the retraining process.

Future work includes several directions. First, we will address the validity threats discussed in Section V to further strengthen the robustness of our findings. Second, it is important to investigate other quality measures proposed in fairness testing, such as naturalness, to assess whether they provide more consistent and meaningful evaluation criteria. Third, exploring the combined effects of multiple quality metrics remains a promising direction. For example, integrating validity with other properties of IDIs may lead to more reliable and comprehensive evaluation frameworks. Additionally, developing fairness testing algorithms that explicitly account for such composite metrics would be an important step forward.

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APPENDIX

A. Data Validity

To address this issue, the notion of *data validity* has been introduced [2]. Data validity can be formalized as a set of constraints derived from the training dataset, which specify whether a given data instance is plausible.

a) t-way Interaction: Let $A = \{A_0, A_1, \ldots, A_{k-1}\}$ be a set of attributes. A t-*way interaction* is defined as a value assignment to a subset of attributes $\hat{A} \subseteq A$ such that $|\hat{A}| = t$. For example, a 2-way interaction can be represented as:

$$\tau = \{(A_i = v_i), (A_j = v_j)\}.$$

b) Validity Constraint Inference: Following [2], data validity is inferred from the training dataset by analyzing the frequency of t-way interactions.

Let $\mathcal{T}_t$ denote the set of all possible t-way interactions. For each interaction $\tau \in \mathcal{T}_t$, we compute its frequency in the training dataset. An interaction is regarded as *invalid* if its frequency is less than a threshold θ.

Formally, the set of validity constraints Φ is defined as:

$$\Phi = \{\neg\tau \mid \mathrm{freq}(\tau) < \theta\}.$$

c) Instance-level Validity: A data instance I is considered *valid* if it satisfies all validity constraints:

$$\forall \phi \in \Phi,\ I \models \phi.$$

Otherwise, the instance is considered *invalid*.

d) Discretization: For numerical attributes, values are discretized into k bins before extracting t-way interactions. This discretization enables the uniform handling of mixed-type tabular data.

Under this formulation, the notion of data validity is parameterized by three hyperparameters: interaction strength t, frequency threshold θ, and the number of bins k. These parameters control the granularity and strictness of validity constraints.

B. Partial Correlation

Partial correlation measures the strength of the linear association between two variables while controlling for the influence of additional variables. Unlike Pearson's correlation coefficient, which reflects both direct and indirect associations, partial correlation isolates the relationship between the variables of interest by statistically removing the effects of confounding factors. This property makes it particularly suitable for analyzing relationships in multivariate settings where confounders may induce spurious correlations.

Definition 4 (Partial Correlation Coefficient): Let r_{xy}, r_{xz}, and r_{yz} denote the Pearson correlation coefficients between X and Y, X and Z, and Y and Z, respectively. The partial correlation coefficient between X and Y given Z, denoted as $r_{xy.z}$, is defined as:

$$r_{xy.z} = \frac{r_{xy} - r_{xz} r_{yz}}{\sqrt{(1 - r_{xz}^2)(1 - r_{yz}^2)}}. \tag{6}$$